Quantitative Data Analysis – Exercises

(Week 09)

In these exercises, you will continue preparing the apartment data. In detail, you will learn how to create new variables, merge the data with geocoded addresses and other data and work with pivot tables. In the data analysis process model, these exercises cover part of the step "Preparing & storing" data (see figure 1). Results of the exercises must be uploaded as separate files (no .zip files) by each student on Moodle. Details on how to submit the results can be found in the tasks below.

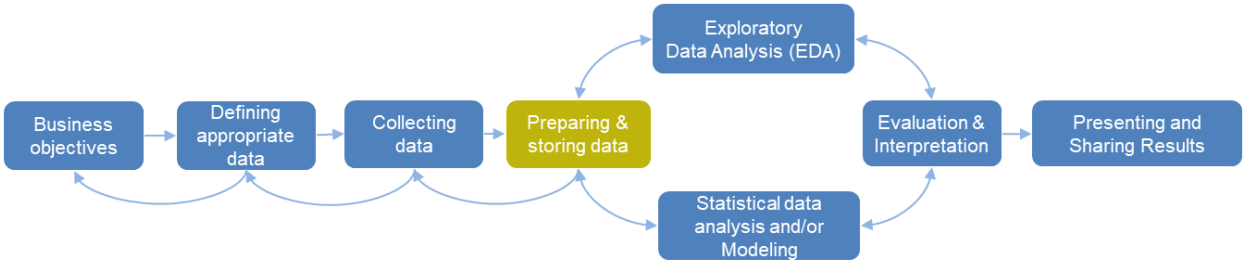


Figure 1: Data analysis process model (see slides of week 07)

Task 1

In these exercises, you will learn to create new variables. Therefore, we will use the rental apartment data of the canton of Zurich ('apartments_data_zuerich.csv'). The tasks are:

1. Run the Jupyter notebook 'apartments_data_preparation_zuerich.ipynb' step by step.
2. In the Jupyter notebook, go to the section 'Create additional variables from the apartment's descriptions' and look at the example under 'Create new binary (0/1) variable 'luxurious''.
3. Based on this example, create the following binary (0/1) variables '**furnished**', '**balcony**', and '**central**'. You are free to create additional variables. Note that the apartment descriptions are mostly written in German (some in English), so the patterns should include German (and if useful English) words. Because the Jupyter notebook converts the descriptions to uppercase (.str.upper()), the patterns must be written in UPPERCASE, e.g. 'MÖBLIERT|FURNISHED'. You can use https://www.deepl.com for translation. State your patterns in the Jupyter notebook, check some matched and unmatched descriptions and look out for negations (e.g. 'UNMÖBLIERT' also contains 'MÖBLIERT').
4. In the Jupyter notebook, go to the example: 'Create new categorical variable based on apartment area'.
5. Based on this example, create a new categorical variable based on the variable 'price_per_m2'. The variable should contain three levels 'low', 'medium', 'high'.

Tip: use e.g. the 33% and 67% quantiles of 'price_per_m2' as thresholds and pd.cut() with bins from 0 to np.inf and right=False as in the example.

6. The output of the Jupyter notebook is a file 'apartments_data_prepared.csv' which will be used in the Jupyter notebook of the following task 2.

   Expected results: Jupyter notebook as html-file 'apartments_data_preparation_zuerich.html' with the additional variables described above.

Task 2

In these exercises you will learn how to merge the data with geocoded data and municipality-level data. The tasks are:

a) Run the Jupyter notebook 'combining_and_organizing_data.ipynb' step by step.
b) Open the file 'municipality_data_with_taxable_income.xlsx'. It contains municipality-level data which are merged with the apartment data in the Jupyter notebook. Look at the details of these data.
c) In the Jupyter notebook, look at the section 'Join municipality data to rental apartment data using .merge()'. In this section, the municipality-level data are merged to the apartments data. These data are:

Table 1: Variables in the file 'municipality_data_with_taxable_income.xlsx'

Variable name	Description of variables
bfs_number	Municipality ID
bfs_name	Municipality name
pop	Number of residents
pop_dens	Population density per km^2
frg_pct	Percentage of foreigners
emp	Number of employees
mean_taxable_income	Mean taxable income per household

Expected results: see Task 3

Task 3

In these exercises, you will learn to work with pivot tables in Python. The tasks are:

a) Use the Jupyter notebook 'combining_and_organizing_data.ipynb' prepared in task 2.
b) Go to the section 'Pivoting data using .pivot_table()' and look at the example.
c) Add the variable `price_per_m2` to the existing pivot table.
d) In the pivot table, compare the `price` with the `price_per_m2`.
e) Is there a relationship between the variable area and the price_per_m2? If so, how can this be explained? Include a short explanation in the Jupyter notebook.

f) Create a new pivot table and use aggfunc='count' as the aggregation function (count a column without missing values, e.g. 'web-scraper-order'). How is the number of apartments distributed in the pivot table? Include a short description in the Jupyter notebook cell.

g) Create a new pivot table with the mean values of `rooms`, `area`, `price`, and `price_per_m2` per municipality. Sort the pivot table by price and area with the most expensive municipalities on top.

Expected results: Jupyter notebook as html-file 'combining_and_organizing_data.html' with the pivot tables and explanations from task 3.